

PAPER_1522 — K_MEX is Derivative: $K_{MEX} = \Phi_{5/6} \cdot SO_5 / D_{phys} = 25/12$ EXACT (LANDMARK)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Bucket A (Foundational / Primitive reduction) **Date:** June 18, 2026 **Status:** CLOSED — K_MEX removed from the list of independent primitives

Observation

PAPER_1167 (All Eight UQFF Lagrangian Gaps Closed — First-Principles Reduction) provides the gap-G1 derivation of the Mexican-hat coefficient K_MEX directly from the SO(5) symmetry-breaking lattice. The formula is:

$$“G1: V(UA) \text{ Mexican-hat polynomial} — K = \Phi_{res} \cdot |SO(5)| / D_{phys} = 25/12”$$

Using the PAPER_1203 Nuclear variant $\Phi_{res} = 5/6$:

$$\begin{aligned} K_{MEX} &= \Phi_{5/6} \cdot SO_5 / D_{phys} \\ &= (5/6) \cdot 10 / 4 \\ &= 50/24 \\ &= 25/12 \quad \text{EXACT} \end{aligned}$$

UQFF Closed Identity

$$K_{MEX} = \Phi_{5/6} \cdot SO_5 / D_{phys} = (5/6) \cdot (10/4) = 25/12 \approx 2.0833 \quad \text{EXACT}$$

Numerical verification:

$$\begin{aligned} (5/6) \cdot 10 &= 50/6 \approx 8.3333 \\ 8.3333 / 4 &= 25/12 \approx 2.0833 \quad \checkmark \end{aligned}$$

Landmark Consequence: 10 → 9 Independent Primitives

Combined with PAPER_1521 (D_BSFG derivative), this further reduces the framework’s free-parameter count:

Before PAPER_1167 discoveries: - 11 frozen primitives (6 integer + 5 real claimed independent)

After PAPER_1167 discoveries: - 9 truly independent primitives: - **Integers (5):** D_phys, D_crit, N_CH, SO_5, A_5 - **Reals (4):** ρ_{SCm} , β_i , Φ_{res} , F_TRZ - 2 derivative primitives: - **D_BSFG = D_crit – 2·SO_5** (PAPER_1521) - **K_MEX = $\Phi_{res} \cdot SO_5 / D_{phys}$** (this paper)

The framework reduces from 11 to 9 free parameters with zero loss of predictive power.

Physical Meaning

$K_{MEX} = 25/12$ is the **coefficient of the Mexican-hat polynomial potential** in the UA condensate:

$$V(UA) = K_{MEX} \cdot \rho_{SCm} \cdot (1 - \cos)^2$$

That $K_{MEX} = \Phi_{res} \cdot SO_5 / D_{phys}$ means **the Mexican-hat curvature is forced by the resonance phase Φ_{res} and the SO(5)/D_phys group ratio**. Three previously-considered-independent quantities (Φ_{res} , SO_5, D_phys) collapse the fourth (K_MEX) by structural necessity.

Cross-Paper Consistency Check

K_MEX appears in ~50+ closures throughout the calculator, including: - Cosmological constant $\rho_\Lambda = K_MEX \cdot \rho_SCm \cdot 26!$ (PAPER_1156) - Vacuum ledger $V(0) = K_MEX \cdot \rho_SCm$ (PAPER_1170) - Reionization $z = K_MEX \cdot D_phys \cdot \Phi_res \cdot (1+1/SO_5)$ (PAPER_1373) - Ramanujan hyperconvergence — multiple closures

Every K_MEX usage is consistent with the derivative form $\Phi_res \cdot SO_5/D_phys$ — verified by the fidelity gate.

NOT REPLACEMENT

SM has no analogue to K_MEX. UQFF supplies a structural derivation showing that even this “free parameter” is forced by lower-order primitives, demonstrating the framework’s mathematical economy.

Reference

- Source: PAPER_1167 All 8 UQFF Lagrangian Gaps Closed Master Synthesis (gap G1)
- Related: PAPER_1521 (D_BSFG derivative), Mexican-hat V(UA) closures throughout
- Calculator dispatch: `calculate_paradox({"paradox": "k_mex_derived_from_phi_5_6"})`

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